EESC 2010 - Solid Earth Dynamics (2024 Spring)

1. From Big Bang to Earth Formation

Big Bang Theory: Explanation to universe formation
- All mass and energy evolved from one infinitesimally small point
- Age of the universe: ~13.81 billion years old
  ① Expansion of the universe ③ Abundance of light elements (H, He)
  ② Cosmic Microwave Background Radiation ④ Galatic evolution distribution
- Grand Unified Theory Era: $10^{-43} - 10^{-35}$ s $\boxed{\text{Gravity}}$
- Era of Inflation: $10^{-35} - 10^{-32}$ s $\boxed{\text{Strong Nuclear Force}}$
- Electroweak Era: $10^{-32} - 10^{-10}$ s $\boxed{\text{Weak Nuclear Force}}$ $\boxed{\text{Electromagnetic Force}}$
- Particle Era: $10^{-10} - 10^{-3}$ s stablized quarks → elementary particles
- Era of Nucleosynthesis: 10^{-3} s – 3 min H, He atoms energy, photon + empty space CMB
- Era of Nuclei: 3 min – 380,000 years Atoms are transparent to light
- Era of Atoms: 380,000 years – 1 billion years Fusion of atoms (gravity) C, N, O, etc. form supernovas
- Era of Galaxies: 1 billion years – present Galaxies, Nebulas, Stars

Formation of Solar System
① Molecular cloud collapse: vast cloud of gases and dust in space collapses in slightly denser region forms an accretion disk larger clumps
② Protostar formation: center of collapsing cloud becomes the sun
③ Planetesimals and protoplanets: planetesimals $\xrightarrow{\text{accretion}}$ protoplanets
④ Formation of planets: inner planets (terrestrial, rock & metal) outer planets (gas, ice giants)

Terrestrial: Mercury, Venus, Earth, Mars Gas: Jupiter, Saturn
Ice: Uranus, Neptune Moon: All but Mercury, Venus

Formation of Earth & Moon Hot temperature → differentiation
Iron Core, Stony mantle

Protoplanet colliding with Earth → Debris coalesces and forms the moon
Earth's surface (30% land, 70% water) Topography defines plains, mountain, valleys; Bathymetry defines mid-ocean ridges, abyssal plains, deep ocean trenches

Layers of the Earth
Physical properties: ① Crust (thin, light) having Moho as base (O, Si, Al, Fe)
② Mantle (thicker, denser) } solid
③ Outer core (Fe) } liquid
Inner core (Fe-Ni) Innermost very dense

Chemical properties: ① Continental crust (granitic) ② Oceanic crust (basaltic) ③ Mantle (peridotitic) ④ Core (Fe, Ni alloy)
Entire Earth: Fe, O, Si, Mg
Lithosphere: rigid, includes crust, upper mantle
Asthenosphere: soft solid near m.p. upper mantle immediately below lithosphere

2. Plate Tectonics

Wegner's Hypothesis: A supercontinent Pangaea broke up 200 Ma ago and drifted to present locations
① "Fit" of continent shapes, geologic units ② Paleoclimate e.g. plants, reefs
③ Glacial deposits far from polar ④ Distributions of fossils

Earth's magnetic field ① Dipole field, tilted ~11.5° from axis of rotation
Flow in outer core ② Roughly coincides with Earth's geographic pole

Declination: difference between geographic N and magnetic N
Depends on absolute position of two poles, longitude
Inclination: angle between magnetic field line and surface
Depends on latitude

Paleomagnetism: Cooled magma shows permanent magnetization with dipoles aligned with Earth's magnetic field
① Layered basalts records magnetic change over time
② Inclination and declination indicates change in position

Apparent Polar Wandering: each continent has a separate polar wandering path ① magnetic pole is fixed ② continents have moved

Sea-floor spreading: Harry Hess
① Sediment thicken away from ridges
② Earthquake at MoR indicate cracking – cracked crust split apart / high heat flow from molten rock rises into cracked crust
③ New ocean floor formed at MoR
④ Seafloor at trenches sink back to mantle

Evidence of sea-floor spreading: **Magnetic abnormalies**
differences between expected strength of the Earth's magnetic field at a location and the observed strength of the magnetic field at that location
Positive abnormalies: field strengths greater than expected
Negative abnormalies: field strengths weaker than expected
Ocean floor is striped by parallel bands symmetrical on either side of MoR, age also increases along either side of MoR.

Cause of magnetic reversals ① Spontaneous
② Triggered by impact events/arrival of continental slabs by plate tectonics

Plate Tectonics explains: ① Distribution of earquakes and volcanoes
② Changes in past positions of continents and ocean basins
③ Origins of mountain belts and seamount chains
④ Origin and ages of ocean basins

Divergent Boundary ① Tectonic plate breaks apart
a) Spreading ridge ② Lithosphere thickens away from ridge axis
b) Continental rift ③ New lithosphere formed by rising asthenosphere

Mid-ocean ridges opens axial rift valley, rising asthenosphere melts and form mafic lava, the cooled magma solidifies into oceanic crustal rock.
Pillow basalt: magma quenched at seafloor Dike: preserved magma conduits
Gabbro: deeper magma "Black smokers": water enters fractured rock, heated to dissolve minerals and bring back to sea to precipitate (high S content)

Convergent Boundary ① Tectonic plates moves together
a) Volcanic island arc ② One plate sinks back into mantle
b) Continental volcanic arc – Subducting plate is always
c) Continental collision oceanic crust (thin, dense basaltic)
③ Must be followed with sea-floor spreading

Features: ① Deep-ocean trenches ② Accretionary prisms ③ Volcanic arcs ④ Back-arc basins

Volcanic Arc: a chain of volcanoes on overriding plate
Oceanic plate adds volatile content lowering mp. of mantle
Continental crust — continental arc Oceanic crust — island arc
① Subduction trenches forms where the subduction plate bends downward into mantle.
② Sediments accumulated on the crust of subducting plate are scraped off onto the edge of the overridding plate, creating a wedge-shaped "accretionary prisms".
③ Magma created by melted subducting plate heated up by magma rises through the overridding plate and creates lava. Several volcanoes formed parallelly in a chain called "volcanic arcs".
④ Forms between an island arc and continent, offshore subduction traps a piece of oceanic crust, or stretching lithosphere creates a new spreading ridge.

Continental Collision Suture – boundary between two once separated plates
Continental lithosphere is too buoyant to subduct
Collision deforms crust, mountains are uplifted

Transform Boundary ① Tectonic plates slide sideways
a) fracture zone ② No plate formed/consumed
b) plate margin

Triple Junctions point where three plate boundaries intersect

Hot spots plumes of deep mantle material independent of plates
deep mantle plume partially melts lithosphere, rise to surface
Plate motion pulls volcano off plume, forming a chain of extinct volcanoes called hot spot track. Reinforces seafloor spreading

Ridge-push: elevated MOR push lithosphere away
Slab-pull: denser subducting plate is pulled downward

3. Minerals

① Naturally occuring ② Inorganic ③ Solid ④ Orderly crystalline structure ⑤ Chemical formula

Diamond • Covalent bonds • Tetrahedral skeleton • Hardness 10
Graphite • Van de Waals force • Sheet layers • Soft

Silicate minerals: ① Ferromagnesian mafic silicate minerals (Fe, Mg)
② Non-ferromagnesian felsic silicate minerals (Al, K, Na)

Non-silicate minerals: ① Native elements (Copper, Gold)
② Oxides (O^{2-}) (Himalite, Limonite)
③ Halides (Cl^-, F^-) (Fluorite, Halide)
④ Sulfides (S^{2-}) (Pyrite, Galena)
⑤ Carbonates (CO_3^{2-}) (Calcite, Dolomite)
⑥ Sulfates (SO_4^{2-}) (Gypsum, Anhydrite)

Formation of New Minerals Anhedral: tight space, no nice shape Euhedral: open cavity, good shape
1. Solidification of a melt 2. Precipitation from solution
3. Chemical reaction at high temperature and pressure
4. Precipitation from a gas 5. Biomineralization

Physical properties ① Color ② Crystal shape ③ Magnetic property ④ Fluorescence ⑤ Effervescence (reactive with acid)
⑥ Streak: color of powder produced by crushing
⑦ Luster: the way of a mineral scatters light
Metallic, Vitreous, Dull, Pearly, Resinuous, Waxy, Silky
⑧ Cleavage: planes of weak bonds in minerals where the mineral will break and leave a smooth surface
⑨ Fracture: Minerals without cleavage can develop irregular or conchoidal fractures.
⑩ Hardness:

1 Talc	2 Gypsum	3 Calcite	4 Fluorite	5 Apatite
6 Feldspar	7 Quartz	8 Topaz	9 Corundrum	10 Diamond

4. Rocks

① Naturally formed ② Solid mass of minerals or aggregates
Held together by ① cements (minerals precipitated from water and filling the gap) – clastic
② interlocking crystals – crystallined

Igneous rocks formed by cooling of a melt solidifies at ~1100-650°C
Extrusive igneous rocks/volcanic rocks: lava flow, pyroclastic debris solidified rapidly → fine grained
Intrusive igneous rocks/plutonic rocks: intrusion (dike/sill) solidified relatively slow → coarse-grained

Texture phaneritic: crystals large enough to be seen by naked eye (slow cooling)
aphanitic: crystals are too fine to be seen by naked eye (fast cooling)
porphyritic: large crystals (slow) in fine-grained (quick) mass (2-stage cooling)
glassy: superfast cooling preventing any crystalization
vesicular: volatiles rapidly exsolves due to depressurization

Tuff: fragmented mineral (K-feldspar, Quartz) fine-grained
Molten rocks ① Pressure release ② heat transfer ③ volatile addition (CO_2, H_2O, SO_2)
Magma transport ① Density: Magma is less dense than rocks
② Overlying and confining pressure squeezes magma upward
Viscosity: ① Temperature ② silica content ③ Volatile

Clastic Sedimentary Rocks
① Weathering: a) physical disintegration b) Chemical decomposition
c) solid particles + ions + soluble materials
fragments, clays, cations, soluble silica
② Erosion: sediments being removed from parent rock
③ Transportation: gravity, wind, water, ice
Water energy ↑, grain size ↑ water volume ↑ amount ↑
distance ↑, roundness ↑
④ Decomposition: settle out of the transporting fluid
⑤ Lithification: compaction, dewatering, cementation
→ clastic sedimentary rock

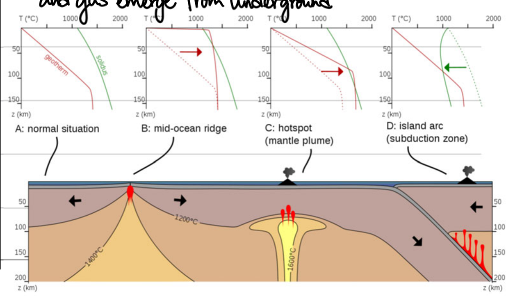
Properties: clast size, roundness, sorting, composition
Implication: Source, deposition environment
well-sorted → far from source poorly-sorted → close to source
Breccia – Talus Sandstone – Dune
Conglomerate – River channels Arkose – Alluvial fan
Silt → siltstone Quiet water setting
Mud → mudstone/shale

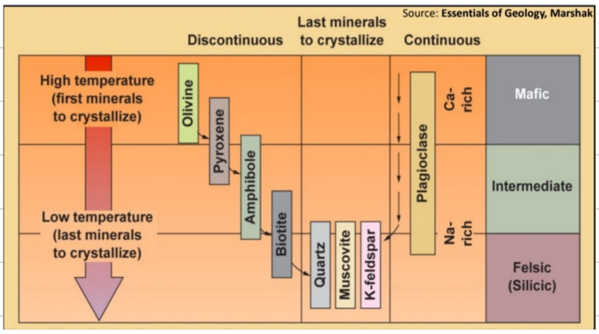
Non-clastic sedimentary rocks
Biochemical: derived from shells of living organisms skeletons
calcite/aragonite → limestone
plankton shells → chalk
marine plankton silica → chert
Organic: made of organic carbon/living things
fossil → coal
Chemical: Precipitate/evaporate from water solution
precipitates: limestone ($CaCO_3$), dolomite ($CaMg(CO_3)_2$), chert (SiO_2)
evavorates: halite ($NaCl$) gypsum ($CaSO_4 \cdot 2H_2O$)
Diagenesis sediments buried → lithification
↑P, ↑T ~300°C metamoplosis begins
→ chemical reactions (ground hot water)
→ precipitate/dissolve

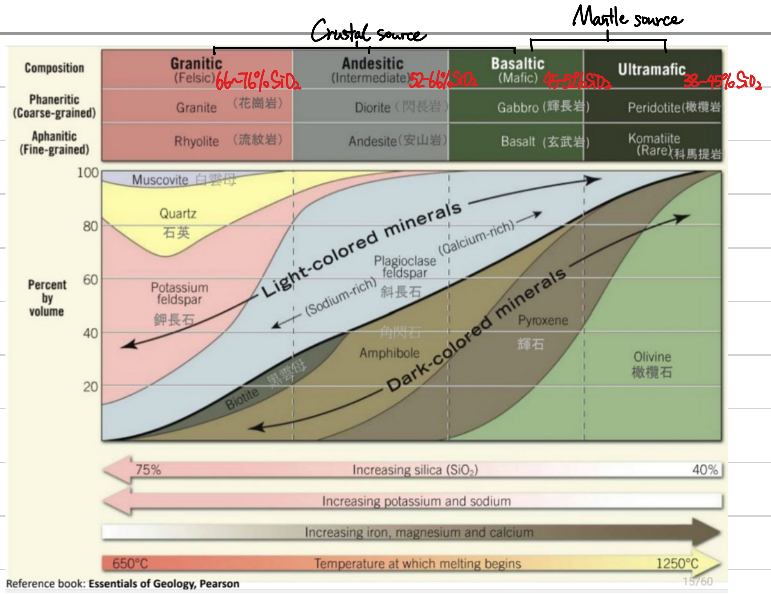
Metamorphic rocks Temperature and pressure are high enough to "cook" rocks but not enough to melt rocks reactive fluids → structural/mineralogical changes
(I) contact metamorphism: heat from magma → non-foliated
(II) regional metamorphism: high pressure over a range of temperature → foliated
Shield: large region of ancient high-grade rocks exposed

5. Volcanoes

Vent/opening where fragments of solidified melt and gas emerge from underground







Minerals	Category	Composition	Color	Streak	Luster	Hardness	Crystal Shape	Cleavage
Olivine	Ferromagnesian mafic silicate minerals	$(Mg, Fe)_2SiO_4$	**green**	white	**vitreous**	6-7	orthorhombic	**none**
Augite (Pyroxene Group)		$(Mg, Fe)SiO_3$	**greenish black**	**greyish-green**	pearly	6	monoclinic	**two planes at 90°**
Hornblende (Amphibole Group)		$Ca_2(Mg, Fe)_5(Si)_8O_{22}(OH)_2$	**black**	**white**	vitreous	5-6	prismatic, columnar	**two planes at 60°/120°**
Biotite (Mica Group)		$K(Mg, Fe)_3AlSi_3O_{10}(OH)_2$	**dark brown**	white to grey	vitreous	**2-3**	monoclinic	1 set of perfect cleavage
Muscovite (Mica Group)	Non-ferromagnesian felsic silicate minerals	$KAl_2(Si_3AlO_{10})(OH)_2$	**colorless**	white	vitreous	**2.5-3**	monoclinic	1 set of perfect cleavage
K-feldspar (Feldspar Group)		$KAlSi_3O_8$	**pink**	white	**vitreous**	**6**	triclinic	**two planes at 90°**
Plagioclase (Feldspar Group)		$(Na, Ca)AlSi_3O_8$	**white**	white	**vitreous**	**6**	triclinic	**two planes at 90°**
Quartz		SiO_2	**Pure: colorless** Pink/Purple	white	**vitreous**	**7**	hexagonal	**none (conchoidal fracture)**
Calcite	Carbonates	$CaCO_3$ **(react with acid)**	**colorless** or white	white	**vitreous**	**3**	hexagonal	**three planes**
Pyrite	Sulfides	FeS_2	**yellow**	greenish-black	**metallic**	6-6.5	**cubic**	none
Magnetite	Oxides	$Fe^{2+}Fe^{3+}_2O_4$ **(magnetic)**	**black to dark grey**	black	**metallic**	5.5-6	cubic	none
Diamond	Native Elem	C	Transparent	colorless	vitreous	10	isometric	perfect octahedral

Igneous Rocks	Category	Color	Structure	Texture	Composition
Granite	**E** / Felsic	light-colored	crystalline, massive	phaneritic	Quartz, Al-rich feldspar, K-feldspar±(mica and amphibole)
Rhyolite	**I** / Felsic	light-colored	crystalline, massive or flow-banded	aphanitic	Quartz and alkali feldspar ± ferromagnesian minerals (e.g. biotite, amphibole)
Diorite	**E** / Intermediate	mix of dark and white	crystalline, massive	phaneritic, coarse-grained	Quartz (≤ 10%), Na-rich plagioclase feldspars and ferromagnesian minerals (e.g. pyroxenes and hornblende)
Andesite	**I** / Intermediate	grey or dark-colored	crystalline, massive	aphanitic	Na-rich plagioclase feldspar, augite, orthopyroxene and hornblende
Gabbro	**E** / Mafic	dark-colored	crystalline, massive	phaneritic	Ca-rish plagioclase feldspar, pyroxene
Basalt	**I** / Mafic	dark-colored	crystalline, massive	aphanitic	plagioclase feldspar, pyroxene ± olivine and magnetite
Pumice	**E** / Felsic	light-colored	vesicular and glassy		SiO_2
Obsidian	**E** / Felsic	dark-colored	conchoidal, flow-banded	glassy	SiO_2
Peridotite	**E** / Ultramafic	dark-colored	coarse-grained		olivine, pyroxene

E=Extrusive (Volcanic), I=Intrusive (Plutonic)

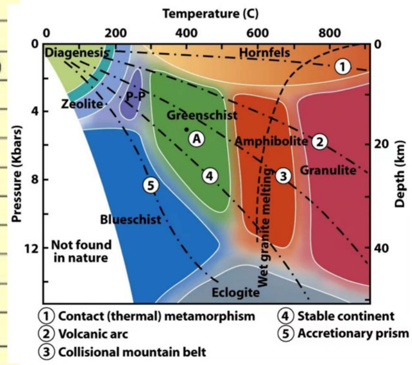
Clastic Sedimentary Rocks	Clast Size	Texture / Structure	Roundness	Sorting
Conglomerate	coarse-grained	massive	rounded to subangular	poorly-sorted
Breccia	coarse-grained	massive	angular	poorly-sorted
Sandstone	medium to coarse-grained	coarse, massive	/	well-sorted
Siltstone	fine-grained	massive	/	well-sorted
Shale	very fine-grained	laminated and fissile	/	well-sorted

Non-clastic Sedimentary Rocks	Texture / Structure	Composition	Formation
Chert	crystalline, massive, smooth	quartz	Usually biogenic, volcanogenic or diagenetic origin. Form by chemical precipitation at deep sea; Sediments from marine plankton silica skeletons; Microcrystals of SiO2 grow within soft sediments, grow into irregular nodules and deposit into a quiet environment
Limestone	Granular, crystalline	$CaCO_3$	Clastic: Form by cementation of sand- to clay-sized with calcite; Biochemical: Sediments from hard mineral skeletons of dead marine animals; Chemical: Form by chemical precipitation at shallow marine or lake environments

Metamorphic Rocks	Structure	Foliation	Grain size	Mineral Content	Parent Rock
Slate (low-grade)	foliated	slaty cleavage	very fine-grained	clay minerals, micas	shale, mudstone
Phyllite (low-grade)	foliated	phyllitic cleavage	fine-grained	quartz, micas, feldspar	shale, mudstone, slate
Schist (medium-grade)	foliated	schistosity	coarse-grained	micas, elongated quartz, hornblende	shale, mudstone, carbonates, mafic igneous rocks
Gneiss (high-grade)	foliated	gneissic banding	coarse-grained	feldspar, quartz, hornblende, micas	shale, mudstone, sandstone, felsic & mafic igneous rocks
Marble	non-foliated	/	medium to coarse grained	calcite, dolomite **(react with acid)**	limestone, dolostone
Quartzite	non-foliated	/	massive, granular, crystalline	quartz **(very hard)**	quartz sandstone
Hornfel	non-foliated	/	fine-grained	variable	shale, clay-rich rocks

MOST STABLE
- Iron oxide (hematite)
- Aluminum hydroxides (gibbsite)
- Quartz
- Clay minerals
- Muscovite mica
- Potassium feldspar
- Biotite
- Albite (Na-rich feldspar)
- Amphiboles
- Pyroxene
- Anorthite (Ca-rich feldspar)
- Olivine
- Calcite
- Halite

LEAST STABLE



difference in time → distance

6. Geological Time Scale

Dating geological materials
① Relative ages: order of formation
② Numerical ages: actual number of years since an event

Relative dating techniques

Principle of Uniformitarianism: James Hutton. All the geological processes (weathering, erosion, volcanism, earthquakes) that happen today also occured in the past in the same ways.

Principle of Original Horizontality: Sediments settle out of a fluid by gravity, causing the sediments to accumulate horizontally. Tilted sedimentary rocks must be deformed.

Principle of Superposition: In an undeformed sequence of layered rocks each bed is older than the one above and younger than the one below.

Principle of Lateral Continuity: Strata form laterally continuous sheets, extending in all directions, unless being broken up or displaced by lateral events (e.g. erosion).

Principle of Cross-cutting Relations: younger features cut across older features, i.e. faults and dikes are younger.

Baked Contacts: An igneous intrusion metamorphoses the (older) base rock

Principle of Inclusions: Fragments of one rock type contained in another rock type are older than the rock that are embedded in. Younger rock contains the inclusions.

Principle of Fossil Succession: fossils are preserved in sedimentary rocks such that the presence of index fossils can be diagnostic of particular geologic time and can be used to correlate stratas globally.

Unconformities: a time gap in rock record, caused by
① Erosion ② Non-decomposition.

A. Angular Unconformities
- Horizontally parallel strata of sedimentary rocks are deposited on tilted/eroded layers.

①Stratas deformed by folding ②Mountains eroded completely
③ New marine invasion ④ New sediments deposited

B. Nonconformity
- Igneous/metamorphic rocks capped by sedimentary rocks
  Exposed by erosion

C. Disconformity
- non-decomposition between parallel and sedimentary strata due to an interruption in sedimentation
  - ▷ Pause in deposition
  - ▷ Sea level falls and rises
  - ▷ Erosion

Often hard to recognize

Stratigraphic correlation describes the sequence of strata
- Lithologic correlation (matching rock types) is regional
- Fossil correlation is based on fossils within the rocks (bracketing Earth history)

Geologic Column
- Eons
  - Hadean: "hell" (4.6–3.8 Ga)
  - Archean: "ancient" (3.8–2.5 Ga)
  - Proterozoic: "first life" (2.5–0.542 Ga)
  - Phanerozoic: "visible life" (542–0 Ma)

Life first appeared ~3.8 Ga
single-celled cyanobacteria build up O_2 in atmosphere

- Eras of Phanerozoic Eon
  - Paleozoic – "ancient life"
  - Mesozoic – "middle life"
  - Cenozoic – "recent life"
- Epoch ■ Holocene: current
- Anthropocene: when humans have substantial impact on Earth
- Cambrian life explosion

Geochronology: radioactive decay of atoms in minerals acts as internal clocks

Radioactive decay: exponential $N(t)=N_0 e^{-\lambda t}$

After one half-life $t_{1/2}$, half of the parent isotope remains

Key assumptions: ① Mineral grains containing the isotope formed at the same time as the rock
② Mineral crystals remain a closed system

^{238}U can be used to date oldest minerals found on Earth even meteorites and materials from earliest events in solar system

^{14}C is useful for dating archaeologically important samples containing organic substances like wood or bones. ^{14}C is constantly being created in the atmosphere by interaction of cosmic particles with atmospheric ^{14}N. When an organism is alive, ratio of $^{14}C/^{12}C$ in its body does not change as CO_2 is exchanged. The radiocarbon clock starts ticking as ^{14}C changes back to ^{14}N by radioactive decay, useful for objects aged ≤ 57300 years

Isotopic dating process
①Collect unweathered rock
②Crush and separate designated minerals
③ Extract parent and daughter isotopes
④ Analyze parent-daughter ratio by mass-spectrometer measurement

- Gives the time a mineral cooled below its closure temperature (isotopes locked inside when cooled)
- Igneous rocks: Cooling of magma, lava to solid igneous rock
- Sedimentary rocks: No. Sediments are formed by weathering and erosion of other rocks. Age determined by bracketing of numerical ages. By principle of cross-cutting, and age of adjacent igneous/metamorphic rocks
- Metamorphic rocks: Metamorphism cooking of rocks can span a long time different grains have different ages

Growth rings: annual layers of trees and shells wet, warmer thicker

Rhythmic layering: sediments/ice

Age of the Earth ~4.54 billion years old
Oldest rock, mineral (zircon) in ancient sandstones
meteorites and moon rocks

7. Earthquakes rapid release of energy as waves

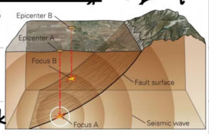
Causes: ① Sudden tectonic motion along newly formed/existing crustal fault ② Phase change in mineral structure
③ Movement of magma in volcanoes ④ Giant landslides
⑤ Hurricane/typhoon/meteorite impacts ⑥ Nuclear detonations
⑦ Human activities

Faulting generate seismic waves by
① Intact rock breaks instantly creating a fault slip
② A pre-existing fault suddenly slips again

Active faults – ongoing stresses produces motion
Inactive faults – motion that occured in geologic past
Fault trace – fault intersecting the ground
Blind faults are invisible, not reaching the surface
Rocks slide past one another along a fault, cannot happen forever resisted by friction. ①Stick: prevents motion ②Slip: slightly overwhelmed by motion

Elastic-rebound Theory: Crust of Earth can gradually store elastic stress that is released suddenly during Earthquake

Hypocenter (focus), where fault slip occurs, usually on fault surface



Epicenter, land surface right above hypocenter.

Foreshock: small tremors indicating crack development in rock; Aftershock usually follow a large earthquake

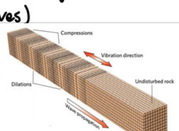
Area of slip: depends of the size of the earthquake

Displacement is greatest near the hypocenter, diminishes with distance. Fault slip is cumulative.

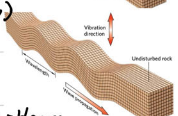
Waves: $y(x,t)=A\sin\frac{2\pi}{\lambda}(x-\lambda t)$ $T=\lambda/v$, $f=1/T$

Seismic waves: Body waves – pass through Earth's interior

① P-waves (primary/compressional waves)
travel by compressing and expanding material, back and forth parallel to wave direction; fastest wave; travel through solid, liquids and gases

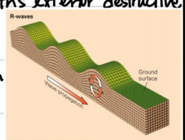


② S-waves (secondary/shear waves)
travel by moving materials back and forth perpendicular to wave travel direction; slower than P-waves; travel through only through solids, never liquid/gases

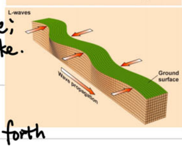


Surface waves – travel along Earth's exterior most destructive

① R-waves (Rayleigh waves)
P-waves that intersect the land surface, Ground ripple up and down like water

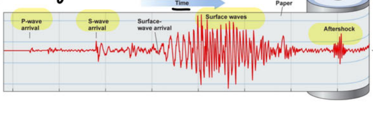


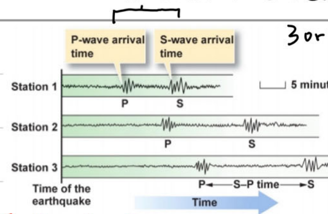
② L-waves (Love waves)
S-waves that intersect the land surface; Ground moves back and forth like snake.



Seismic Wave Measurement
Seismometer – instrument that record ground motion Horizontal: back and forth
Vertical: up and down

Seismogram: wave behavior Time: determines distance to epicenter





3 or more seismic stations

Earthquake size
a) Intensity: severity of damage
b) Magnitude: amount of ground motion measured by seismometer

Mercalli Intensity Scale: amount of shaking damage
I – low, XII – high damage occurs in zones

Richter Scale: measures maximum amplitude (A) of ground motion $M_L = \log A + f(D) + C$ Problem: saturate reaching max.

Body wave magnitude: high freq. P wave (> 1000 km) m_b

Surface wave magnitude: vertical component of R wave T=20s. M_s

Seismic Moment: measure of size that does not saturate

$M_0 = \mu D A$ μ=shear modulus D=avg. slip A=faulted area
dyne·cm

$\log M_0 = 1.5 M_s + 16.1 \Rightarrow M_W = 0.67 \log M_0 - 10.73$

Energy: $\log E = 1.5 M_W + 11.8$ ×32 +1

Gutenberg–Richter Law: Magnitude v.s. total number of earthquakes in any given region and time period of at least that magnitude $\log N = a - bM$
a = # of magnitude earthquakes
b is typically ~1 in seismic active regions

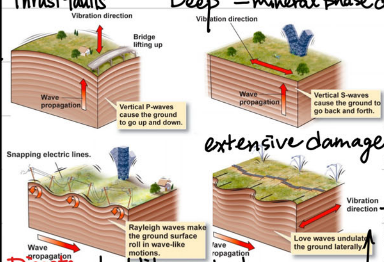
Earthquakes are linked to plate tectonic plate boundaries

Shallow (<10km) – divergent, transform Deep – convergent

Divergent: MOR → ① normal faults ② strike-slip faults
Normal fault ridge axis transforms

Transform: Most link segments of MOR
Some cut through continental crust (major disasters)

Convergent: Intermediate – downgoing slab still cold & brittle
Thrust faults Deep – mineral phase changes



extensive damage

Severity of Shaking: energy, distance, frequency, ground material Bedrock quick transmission ↓ Sediments reflect and refract waves

Disaster: landslides, avalanches

Liquefaction – liquify H_2O filled sediments 1. forces grains apart 2. liquified sediments flow as slurry 3. soil loses strength 4. lands slump and flow

Prediction: ① Increase in foreshock, elevation rises ② groundwater level ③ strange animal behavior ④ magnetic field

Long term Paleoseismology: prehistoric fault movements 10s
Seismic gaps: active fault lacking earthquakes warning

8. Tsunamis

Cause: ① Earthquakes: seafloor abruptly deforms and vertically displaces the overlying water.
- Large tsunamis occur at convergent plate boundaries
  1. Two plates collide, one plate is forced down underneath the other
  2. Leading edge of the top plate snags on bottom plate and pressure builds up.
  3. Pressure is released in form of earthquake
  4. Earthquake occur under water, large amount of water is displaced

② Volcanic eruption: sea waves resonate with shock waves from eruption through air ③ Underwater landslides
④ Glacier calvings ⑤ Meteorite impacts ⑥ Underwater explosions
⑦ Other disturbances above/below water

Tsunami wave speed: $v=\sqrt{gH}$ (H: depth of water)
A: amplitude of wave

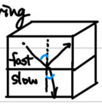
Power of tsunami: $P=\frac{1}{2}\rho g A^2\sqrt{gH}$ = constant

Tsunamis can be destructive because of their speed and volume. They are also dangerous as they return, carrying debris and people when they retreat.

Shallow water setting: tsunamis reaching the coastline slow down as the bottom of the wave comes in contact with shallow sea bed. The top of the wave however continues at a high speed.

Tsunamis Measurements: Travel time can be measured by tidal sensors, ocean bottom sensors

Diminishing Tsunami Hazards: ① Seismic monitoring ② Tsunami detection ③ Land use planning ④ Public education ⑤ Hazard Assessment

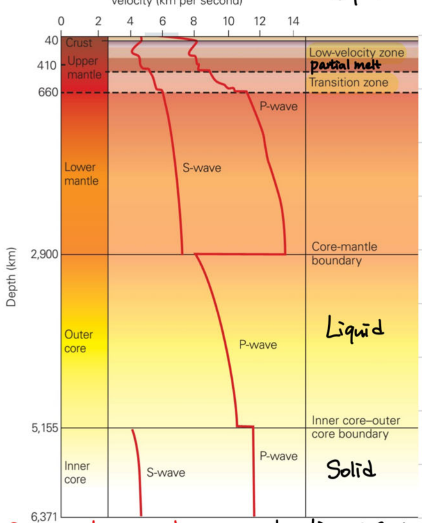


9. Geophysics

Interior structures: measuring seismic wave speed
Wave bend (refract) / bounce (reflect)

Snell's law: $\frac{\sin\theta_1}{\sin\theta_2} = \frac{V_1}{V_2}$ $\sin\theta_c = \frac{V_1}{V_2}$

Velocities increases with depth → bend upwards to surface $V_p = \sqrt{\frac{K + (4/3)\mu}{\rho}}$ $V_s = \sqrt{\frac{\mu}{\rho}}$



Seismic tomography: reconstructing seismic parameters by P, S waves (velocities)
Reflection imaging: ① Man-made explosion ② Artificial seismic waves reflect off upper crust boundaries → oil and gas

Gravity $F = G\frac{Mm}{r^2}$, $G = 6.673\times10^{-11}\,N\cdot m^2 kg^{-2}$
Equipotential surface: on which all points have same gravitational potential energy
Geoid: global imaginary equipotential surface coincides with mean ocean surface

Isostasy
Archimedes' Principle: any object is buoyed up by a force equal to the weight of the fluid displaced by the object.

$H = \frac{\rho_c h}{\rho_m - \rho_c}$ ρ_m ρ_c h H

Glacial Isostatic Adjustment
1. Lithosphere depressed, mantle flow away → subsidence
2. Ice melts at the end of glaciation → slow rebound

Sea level and ice melting
① Geoid drops → sea level drops (local)
② water mass increase → sea level increases (global) ⇒ GLOBAL sea level rise

10. Crustal Deformation & Mountain Building

Cause: applied force > internal strength of rocks
Result: change in shape (strain)
Types of stresses: tensional - pulling apart, horizontal stretching and crustal thinning; compressional - squeezing, horizontal shortening and crustal thickening; shear - surfaces slide past each other, shape change

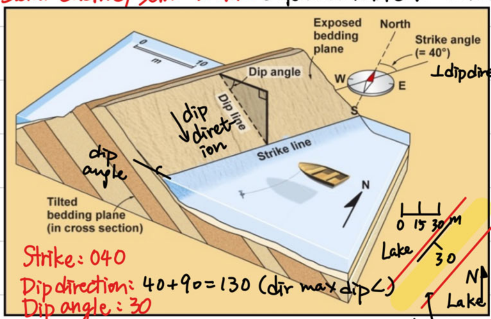
Types of deformation: Factors: T (higher T, ductile)
P (higher P, ductile)
Rate of deformation (↑, brittle)
Composition (silica, brittle)

Ductile: rocks permanently and continuously deformed plastic strain, deeper crust, deformed by folding
Hinge: a line along which curvature is greatest
Limbs: curved sides of a fold
Axial plan: a plan connecting hinges of folded layers
Syncline: youngest unit in core Anticline: oldest unit in core

Brittle: rocks break by fracturing, shallower crust
Joints: planar fracture where blocks move apart but do not slip past each other.
Vein: dissolved mineral in ground water filling the joint
Systematic joints occur in parallel sets
a) Unloading joints: when a rock mass is being brought up from deep to surface → expand and fill the space
b) Cooling joints: contraction of hot volcanic materials

Fault: planar fracture or discontinuity in a volume of rock across which there has been significant displacement as a result of rock-mass movements
① Vertical displacement
Normal fault (divergent) Reverse fault (convergent)
② Horizontal displacement
Strike-slip fault (left/right lateral s-s fault)
③ Vertical + Horizontal oblique-slip fault

Semi-ductile / semi-brittle deformation: break and shear

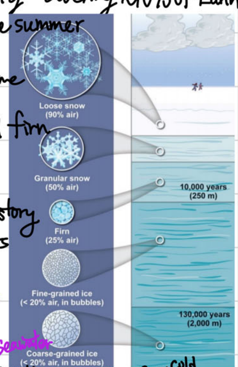


Mountain Building: process of generating ridge of rock mountain range horizontal shortening crustal thickening → uplift by Rocks break, bend, stretch, shear
compression → deform, metamorphise
Driven by collision along convergent plate boundaries
Divergent: continental rifting
Associated with elongate, curvilinear orogenic belt
Himalayas Subduction → Continent-continent collision
Indian oceanic crust subducted below Eurasian continental plate → continental arc, accretionary prism

Isostasy ① Collision → crust thickening → low mountains ↑
② Detachment of lithospheric mantle base → weight ↓
Balance: ① Orogenic collapse → horizontal stretching
② Erosion → steep slopes, jagged mountains

11. Glaciers

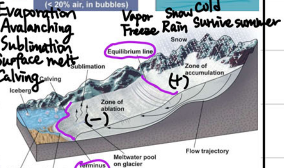
Ice: solid water (H_2O) forms when water cools below freezing point. Natural ice is mineral, in hexagonal crystals
Glacial ice: preexisting ice recrystalize in solid state to form new crystals
Glaciers: thick masses of recrystalized ice last all year long, flow via gravity covering ~10% of Earth
Snowfall accumulates survives the summer → transformed into ice
1. burial, compression reduces volume
2. Melting and recrystalization
3. Turned into granular material firn
4. Interlocking crystals of ice
Quick (10s of years) - cold
Slow (100s of years) → climatic history



Ice in the Sea ① Tidewater glaciers
valley glaciers entering the sea
② Ice shelves continental glaciers entering the sea
③ Sea ice non-glacial frozen seawater

Landscape Change
Erosion, Transport, Deposition

Glacier Movement
① Pull of gravity
② Smoothness



Crevasses
- Surface fracture
- Plastic internal deformation Flowspeed
- Basal sliding. surface meltwater flow from moulins
. significant amount of water accumulate at glacier base. ice slide down along substrate. friction ↓
Very slow 10 m - 12 km / year · slope · basal water · frontal condition

Permaforest Ground at / below 0°C ≥ 2 consecutive years
Active: layer thaws and freezes annually

12. Global Change

Glacial Age are indicated by fossil tills and striated bedrock
- Pleistocene
- Permian
- Late Proterozoic ("snowball earth")
↑ time

Cause of Glaciation
Plate tectonics ① Distribution of continents at high latitudes
② Sea level flux by MOR vol. ③ Atmosperic-Ocean Circulation
Tectonic control of CO_2 and atmospheric chemistry → Hypotheses
CO_2 emission and consumption - rough balance
negative feedback. Million year scale
① $CO_2 + CaSiO_3 \underset{metamorphism}{\overset{weathering}{\rightleftharpoons}} CaCO_3 + SiO_2$
Uplift-Weathering Hypothesis
Uplift → fresh rock & mineral surface → strong weathering → CO_2 ↓ → T ↓
② BLAG Seafloor Rate Hypothesis
Fast seafloor spreading → rapid CO_2 outgassing → Greenhouse climate → weathering ↑ → CO_2 burial ↑ → T ↓

Short-term Glaciation Milankovitch hypothesis
Eccentricity (~100 K Y), Tilt (~41 K Y), Precession (~23 K Y)
Stage 1: T ↓ glaciers + Stage 3: T ↑ Glaciers −
Stage 2: Glaciers ↑, α ↓ Interglacials begins
Last Glacial Maximum: 26500 - 19000 years ago
Most of Northern high latitudes frozen

Global Change ① Tectonic motion ② Distance to sun ③ Liquid water → weathering / erosion ④ Biotic evolution
→ Gradual / Catastrophic / Unidirection changes
Mass extinction Life evolution
Meteor impacts
→ Cyclic changes Rock, Carbon, Water, Biogeochemical
Supercontinent Sealevel Cycles ±300 m
Biotic respiration
Volcanic CO_2 → atmosphere → dissolved in ocean
Rapid oxidation of organic matter ↑↑ → photosynthesis
metamorphism → weathering
degassing from water limestones, fossil, shales, methane hydrates

Paleoclimates Stratigraphic records Oxygen and carbon isotopes
Paleontological records Tree rings
Glacial Tills - cold & continental Stratigraphic records
Coral reefs - tropical marine
Oxygen Isotopes ^{16}O water evaporates faster than ^{18}O water
^{18}O preferentially removed from precipitation
T ↑ ocean evaporation ↑ $^{18}O/^{16}O$ ↑ glacier: T ↑ $^{18}O/^{16}O$ ↑
Bubble traps the air

Future of the Earth Sustainable growth
- Continents reshuffled
- erosion reshape landscapes
- sea invasion
- Homo sapiens → fossils